



Course Title	Object Oriented Programming with C++ and Java	Course Code	BCSESC305A
Course Instructor	Dr. Usha G R, Sheik Imran, Prof. Bhagya S G, Prof. Anusha N, Prof. Priya V	Semester-Section	III
IA Test No.	I	Max. Marks	30
Date	16.11.2025	Time	9.30 – 10.45 AM

Course Outcome Statements: After the successful completion of the course, the students will be able to	
CO1	Understand and apply fundamental C++ object-oriented principles such as classes, objects, constructors, destructors, and function overloading to develop basic programs.
CO2	Utilize C++ templates and Standard Template Library (STL) components—including containers, algorithms, and iterators—to develop efficient and reusable solutions for problem-solving.
CO3	Utilize fundamental Java programming concepts—such as data types, control structures, and object-oriented design—to develop simple and structured applications.
CO4	Utilize advanced Java features such as method overloading, access control, inheritance, and exception handling to develop efficient, reliable, and reusable programs.
CO5	Design and implement well-organized, concurrent Java programs by utilizing packages, interfaces, and multithreading to enhance code modularity and execution efficiency.

Note: Answer any one full question from each Part.					
Q. No.		Question	Marks	RBT Level	CO
Part A					
1	A	What is object-oriented programming? Describe the main features of C++.	6	L2	1
	B	Write a C++ program to compute the area of triangle, square, rectangle and circle by overloading the area () function.	6	L3	1
OR					
2	A	Describe reference variables and demonstrate their usage with an example.	6	L2	1
	B	Develop a C++ program to demonstrate a static member variable and a static member function.	6	L3	1
Part B					
3	A	Explain class template with multiple parameters.	6	L2	2
	B	A debugging utility in a software project requires swapping values of any data type integers, floats, or even characters, without rewriting code for each type. (Use of generic member function template).	6	L3	2
OR					
4	A	Explain the components of STL with an appropriate diagram.	6	L2	2
	B	A teacher wants to sort the marks of students in ascending order. Since the marks can be in integer, float, or double format, write a generic function using template that can sort data of any numeric type.	6	L3	2
Part C					
5	A	List and explain different Java Buzzwords.	6	L2	3
		OR			
6	A	Describe different Integer data types in Java.	6	L2	3

RBT (Revised Bloom's Taxonomy) Levels : Cognitive Domain		
L1 : Remembering	L2 : Understanding	L3 : Applying
L4 : Analysing	L5 : Evaluating	L6 : Creating

Course Coordinator

Coordinator
DAAC

Program Coordinator

BOE Chairman



Bapuji Educational Association®
Bapuji Institute of Engineering and Technology, Davanagere
An Autonomous Institute Affiliated to VTU
Department of CSE/ISE and Allied Branches

Course Title	Object Oriented Programming with C++ and Java	Course Code	BCSESC305A
Course Instructor	Dr. Usha G R, Prof. Sheik Imran, Prof. Bhagya S G, Prof. Anusha N, Prof. Priya V	Semester-Section	III
IA Test No.	II	Max. Marks	30
Date	10.01.2026	Time	9.30 – 10.45 AM

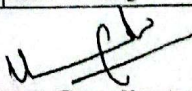
Course Outcome Statements: After the successful completion of the course, the students will be able to	
CO1	Understand and apply fundamental C++ object-oriented principles such as classes, objects, constructors, destructors, and function overloading to develop basic programs.
CO2	Utilize C++ templates and Standard Template Library (STL) components—including containers, algorithms, and iterators—to develop efficient and reusable solutions for problem-solving.
CO3	Utilize fundamental Java programming concepts—such as data types, control structures, and object-oriented design—to develop simple and structured applications.
CO4	Utilize advanced Java features such as method overloading, access control, inheritance, and exception handling to develop efficient, reliable, and reusable programs.
CO5	Design and implement well-organized, concurrent Java programs by utilizing packages, interfaces, and multithreading to enhance code modularity and execution efficiency.

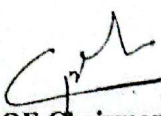
Note: Answer any one full question from each Part.					
Q. No.		Question	Marks	RBT Level	CO
Part A					
1	A	Demonstrate the use of the super keyword in an inheritance program with a suitable example.	6	L2	4
	B	A school is developing a simple calculator program for students. While calculating results, the program must divide total marks by the number of subjects. If the number of subjects is entered as zero , the program should not crash . Instead, it should raise a custom exception called DivisionByZeroException and display a meaningful error message.	6	L3	4
OR					
2	A	Define Inheritance. Develop a Java program to demonstrate multilevel inheritance.	6	L3	4
	B	Explain Multiple catch clauses in Java with an example program.	6	L2	4
Part B					
3	A	Develop a Java program to create a package named mypack and import & implement it in a suitable class.	6	L2	5
	B	Define thread. Explain two ways of creating a thread in java with an example.	6	L2	5
OR					
4	A	Demonstrate the use of Interfaces with an example program.	6	L2	5
	B	Explain the life cycle of a thread with a neat diagram.	6	L2	5
Part C					
5	A	Explain the declaration and initialization of one and two-dimensional arrays with an example for each.	6	L2	3
		OR			
6	A	Explain the syntax of for loop and for each loop with an example	6	L2	3

RBT (Revised Bloom's Taxonomy) Levels : Cognitive Domain		
L1 : Remembering	L2 : Understanding	L3 : Applying
L4 : Analysing	L5 : Evaluating	L6 : Creating


Course Coordinator


Coordinator DAAC


Program Coordinator


BOE Chairman 31/12/2024